

How do you run CS223 labs on a Mac without Vivado?

19 September 2026 · M2, 8 GB, macOS 15 [PDF](#) [the five project files](#) section 1 prerequisites, 2 the one-line install, 3 the setup by hand, 4 to 8 your lab and the limits

Apple Silicon Mac (M1 or later), a Digilent Basys3, free tools. Everything here was run on an M2 with 8 GB of RAM and macOS 15 on 19 and 20 September 2026. Intel Macs are untested and the installer refuses them.

Two ways in. **Take the one line.** Build by hand only if you want to see each tool go in, or the installer refuses your machine.

	one line	by hand
do	section 1, then 2, then 5 to 8	section 1, then 3, then 4 to 8
time after section 1	one command; 3 min 37 s to 4 min 17 s on the tested machine, 1.4 GB downloaded	5 steps; about 5 minutes of compiling plus 1.76 GB of full clones, call it 20 minutes
you get	the dewfpga command	the same tools plus a Makefile you own
tool versions	two pinned commits, Homebrew versions in section 9	the same two commits, typed by you

Section 4 is the Makefile project the by-hand path uses. Sections 5 to 8 are for both: your own lab, VS Code, the one course file that breaks, what this chain cannot do.

If a step fails. Read the last lines in the terminal. Every error hit during this setup is filed under its exact text, with the fix: nosey-dewdrop.github.io/dewfpga/errors. Not there? Open an issue at github.com/nosey-dewdrop/dewfpga/issues with the last ten lines and the section number.

No Mac or no board with you? The [browser simulator](#) runs the same three files (design.sv, tb.sv, the .xdc) with nothing installed: a virtual Basys3, and a testbench view that draws the waveform.

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1. What do you need before you start?

The board. A Digilent Basys3 and its micro-USB cable. Apple Silicon MacBooks have only USB-C ports, so you also need a USB-C to USB-A adapter or hub. The board is powered from the cable; leave the JP2 jumper on USB.

The terminal. Press Cmd Space, type Terminal, press Enter. Every command in this guide goes into that window: paste it, press Enter, wait until the line ending in % comes back before you paste the next one. Some commands span several lines and end in \; copy the whole grey block at once.

Apple's command line tools. They give your Mac git, make and the C compiler.

```
xcode-select --install
```

A window pops up; click Install and wait. If instead the terminal prints xcode-select: note: Command line tools are already installed, that is fine. Check:

```
xcode-select -p
# /Library/Developer/CommandLineTools
```

Homebrew. It installs developer tools by name.

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

It asks for your Mac password; the letters do not show while you type. At the end it prints a few lines under "Next steps". Paste those into the terminal too, or the next command answers command not found. Check:

```
brew --version
# Homebrew 4.x
```

VS Code. Download it from code.visualstudio.com. Then in VS Code press Cmd Shift P, type shell command, choose "Install 'code' command in PATH". After that code . in the terminal opens the current folder. Do not edit these files in TextEdit; it saves rich text, and the tools cannot read that.

2. How do you install it in one line?

```
curl -fsSL https://nosey-dewdrop.github.io/dewfpga/install | bash
```

3 min 37 s to 4 min 17 s over three runs on the tested machine, 1.4 GB downloaded; on slow wifi it takes longer, and it prints progress the whole time. It never asks for a password. It puts the dewfpga command into ~/.dewfpga (its sha256 is compared with nosey–dewdrop.github.io/dewfpga/dewfpga.tgz.sha256, so a broken download stops here), links it into Homebrew's bin, installs Yosys, openFPGALoader, Icarus Verilog and the build tools of section 3.1 through Homebrew, and builds nextpnr-xilinx and prjxray into ~/fpga at the two commits in section 9. Nothing touches your system Python. The last lines it prints:

```
all good.

Total: 217 s. Log: /Users/you/fpga/install.log
Next:  dewfpga new blink && cd blink && dewfpga flash
```

To read the script before running it:

```
curl -fsSL https://nosey-dewdrop.github.io/dewfpga/install -o install.sh
code install.sh      # read it, then:
bash install.sh
```

Check that every piece is there:

```
dewfpga check
```

```
dewfpga check  (FPGA_HOME=/Users/you/fpga)
✓ yosys          /opt/homebrew/bin/yosys
✓ iverilog       /opt/homebrew/bin/iverilog
✓ openFPGALoader /opt/homebrew/bin/openFPGALoader
✓ nextpnr-xilinx /Users/you/fpga/nextpnr-xilinx/build/nextpnr-xilinx
✓ xc7frames2bit  /Users/you/fpga/prjxray/build/tools/xc7frames2bit
✓ fasm2frames    /Users/you/fpga/prjxray/utils/fasm2frames.py
✓ chipdb         /Users/you/fpga/chipdb/xc7a35t.bin
✓ venv           /Users/you/fpga/venv/bin/python
✓ python pkgs    fasm prjxray pyyaml textx simplejson intervaltree
all good.
```

A red x names the missing piece: run the curl line again (it is the same as dewfpga install), it skips what is done. dewfpga: command not found has two causes: the Homebrew "Next steps" lines from section 1 were not pasted (paste them, open a new terminal window), or the installer stopped before its last line; scroll up to the first ERROR: line and look it up on the errors page.

Now plug the Basys3 into the Mac with the USB cable (the PROG port, next to the power switch) and flip the power switch on. Make the example project and flash it:

```
dewfpga new blink    # prints: blink/ ready: cd blink && dewfpga flash    (VS Code: %
1B)
cd blink
dewfpga sim          # simulation
dewfpga bit          # .sv -> .bit
dewfpga flash        # .bit -> the board; runs bit first when the .bit is missing or o
lder than the .sv
```

What each one prints when it works. `dewfpga sim`:

```
VCD info: dumpfile blink.vcd opened for output.
PASS: 3 checks (see the waveform for the toggle)
blink_tb.sv:35: $finish called at 200000 (1ps)
```

`dewfpga bit`, three lines: the pin check, place and route, the bitstream. The full place-and-route log is in `blink.log`.

```
xdc ok: 33 ports, all mapped.
pnr ok: 73 LUT, 27 FF, 278.71 MHz (PASS at 100.00 MHz)    (full log: blink.log)
blink.bit 2.2 MB
```

`dewfpga flash`:

```
Jtag frequency : requested 6.00MHz    -> real 6.00MHz
Open file DONE
Parse file DONE
load program
Load SRAM: [=====] 100.00%

Done
Shift IR 35
ir: 1 isc_done 1 isc_ena 0 init 1 done 1
```

done 1 means the chip accepted the design. Flip switch 0 up (SW0, the rightmost of the 16 switches): LED 15 (LD15, leftmost) lights and LED 0 (LD0, rightmost, just above SW0) blinks once a second. The design lives in SRAM, so a power cycle clears it; flash again.

If it prints `ERROR: board not found`. Is the USB (PR0G port) plugged in and the power switch ON? instead, the Mac does not see the board: power switch, PROG port, a charge-only cable, the adapter, or the JP2 jumper. The [board not found](#) page goes through them in order.

The rules `dewfpga` follows in a folder:

- Every `.sv` and `.v` in the folder is synthesized, so submodules can live in their own files, and a file can be named anything: `lab5.sv` may hold module `top_design`. A module without ports, or one that calls `$finish`, is a testbench and stays out of the bitstream. One design per folder.
- The top module is the design module that no other module instantiates, as Vivado picks it. When two modules qualify, `dewfpga` names both and you choose: `dewfpga bit lab4` (a module name, or a file name). With several files, `bit` prints the top it chose.

- The pin file is <top>.xdc, or the only .xdc in the folder. Pins in it that the design does not use are ignored, so a fully uncommented Basys3_Master.xdc is fine.
- dewfpfga sim needs a testbench that instantiates the top; <top>_tb.sv is the usual name but any name works. bit and flash do not need one. A testbench that never reaches \$finish is stopped after 120 s with an error; SIM_TIMEOUT=600 dewfpfga sim waits longer. \$stop ends the run like \$finish. flash builds the .bit first if it is missing or older than the sources, so dewfpfga flash alone is enough.
- Two things Vivado lets through and Yosys does not are handled for you: the course's SevenSegmentDisplay.sv port line (output [6:0] seg, logic dp) gets its direction written in, with a note naming the line; an instance without a name (clk_div(clk, out);) and a latch in an always_comb stop with the line and the fix. Section 7 has the details.
- dewfpfga bit writes no bitstream when timing is not met, and a build that fails removes the previous .bit, so flash can never load yesterday's design by mistake. When nothing changed, bit prints one line saying so. \$display in synthesizable code is dropped, as Vivado does. dewfpfga sim exits with an error when the testbench prints \$error or \$fatal, so a red run is visible.
- bit leaves its working files next to the sources (.json, .fasm, .frames, .log, _routed.json), sim leaves <top>_sim, <top>_sim.out and <top>.vcd, flash leaves <top>_flash.out. dewfpfga clean removes all of them and the .bit; the .sv and .xdc stay. Run it before you hand a lab in.

dewfpfga install	install the toolchain (safe to re-run)
dewfpfga check	is every piece in place
dewfpfga sim bit flash clean [top]	work on the .sv files in the current folder
dewfpfga new <dir>	blink example with a VS Code task (Cmd Shift B = flash)
dewfpfga uninstall	remove the tools in ~/fpga, ~/.dewfpfga and the links; brew packages stay

Continue with [section 5](#).

3. How do you build the chain by hand?

These are the commands the installer runs, at the same pinned commits. Vivado does five jobs for a lab; six free tools do them here, two of them for the bitstream:

```

your_design.sv
|-> Icarus Verilog    -> simulation
|-> Yosys             -> synthesis
|-> nextpnr-xilinx    -> place and route
|-> fasm2frames        -> configuration frames
|-> xc7frames2bit      -> your_design.bit
|-> openFPGALoader    -> the board, over USB

```

Three come from Homebrew, two you compile, and fasm2frames is a Python script inside the prjxray checkout. nextpnr also needs a chip database, generated once by another Python script.

Everything lands in ~/fpga: 1.4 GB with the installer's shallow clones, 1.76 GB with the full clones below. The Makefile in section 4 expects exactly this layout: ~/fpga/venv, ~/fpga/nextpnr-xilinx/build, ~/fpga/chipdb/xc7a35t.bin, ~/fpga/prjxray.

3.1 Homebrew packages

```
brew install yosys openfpgaloader icarus-verilog cmake ninja eigen pkg-config python@3.14
```

The first three are tools from the diagram; the rest are what the two builds below need. A few hundred MB. Check:

```
yosys -V                # Yosys 0.69+post
openFPGALoader --Version # v1.1.1
iverilog -V | head -1    # Icarus Verilog version 13.0 (stable)
```

Newer numbers are expected; only the ones in section 9 were tested.

3.2 Python venv

fasm2frames (3.5) and the chip database generator (3.4) are Python scripts. Homebrew's Python refuses a plain `pip install` ([externally-managed-environment](#)), so they get a private one.

```
mkdir -p ~/fpga
"${brew --prefix}/opt/python@3.14/bin/python3.14" -m venv ~/fpga/venv
~/fpga/venv/bin/pip install fasm==0.0.2.post88 pyyaml==6.0.3 textx==4.4.0 simplejson==4.1.2 intervaltree==3.2.1 numpy==2.5.3 pyjson5==2.0.1
```

Check (prints nothing when it works):

```
~/fpga/venv/bin/python -c "import fasm, yaml, textx, simplejson, intervaltree"
```

3.3 nextpnr-xilinx

Place and route. Homebrew has no macOS build, so you compile it. The clone is most of the download; the compile is about 1.5 minutes on this machine.

```
cd ~/fpga
git clone --recursive https://github.com/openXC7/nextpnr-xilinx.git
cd nextpnr-xilinx
git checkout 3fd78784c7788f93f276358edf5477221cc6c179
git submodule update --init --recursive
cmake -B build -G Ninja \
  -DARCH=xilinx -DCMAKE_BUILD_TYPE=Release \
  -DUSE_OPENMP=OFF -DBUILD_GUI=OFF -DBUILD_PYTHON=OFF -DBUILD_TESTS=OFF
ninja -C build -j3 nextpnr-xilinx bbasm
```

-DUSE_OPENMP=OFF is required: Apple's compiler has no OpenMP ([fopenmp](#)). -j3 fits 8 GB of RAM; with 16 GB or more use -j\$(sysctl -n hw.ncpu). Check:

```
~/fpga/nextpnr-xilinx/build/nextpnr-xilinx --version
# "nextpnr-xilinx" -- Next Generation Place and Route (Version 0.9.6)
```

3.4 Chip database

nextpnr needs a description of the XC7A35T. It is not in the download; you generate it once ([no-chipdb](#)). About 35 seconds and 860 MB of RAM while it runs.

```
mkdir -p ~/fpga/chipdb
cd ~/fpga/nextpnr-xilinx
~/fpga/venv/bin/python xilinx/python/bbaexport.py \
  --xray xilinx/external/prjxray-db/artix7 \
  --metadata xilinx/external/nextpnr-xilinx-meta/artix7 \
  --device xc7a35tcpg236-1 \
  --constids xilinx/constids.inc \
  --bba ~/fpga/chipdb/xc7a35t.bba
build/bbasm -l ~/fpga/chipdb/xc7a35t.bba ~/fpga/chipdb/xc7a35t.bin
rm ~/fpga/chipdb/xc7a35t.bba
```

Check:

```
ls -lh ~/fpga/chipdb/xc7a35t.bin
# about 90 MB
```

Another 7-series board would need its own --device. Untested.

3.5 prjxray

Turns nextpnr's text output into the binary .bit: one Python script, one compiled tool. About 2 minutes.

```
cd ~/fpga
git clone --recursive https://github.com/f4pga/prjxray.git
cd prjxray
git checkout c9f02d8576042325425824647ab5555b1bc77833
git submodule update --init --recursive
~/fpga/venv/bin/pip install --no-deps -e .
cmake -B build -DCMAKE_BUILD_TYPE=Release -DCMAKE_POLICY_VERSION_MINIMUM=3.5 -DPRJXRAY_BUILD_TESTING=OFF
cmake --build build --target xc7frames2bit -j3
```

--recursive is required ([yaml-cpp-not-built](#)). -DCMAKE_POLICY_VERSION_MINIMUM=3.5 is required ([cmake-policy-version-minimum](#)). prjxray's own README says to install Vivado 2017.2; that is for mapping a new chip, and ours is mapped. Ignore it. Check:

```
ls ~/fpga/prjxray/build/tools/xc7frames2bit ~/fpga/prjxray/utils/fasm2frames.py
```

Both paths print back. The full check is `make check` in the next section, one line per tool.

4. What is in the project folder?

A project is one folder. Download the files, do not type them:

```
mkdir -p ~/cs223/blink && cd ~/cs223/blink
curl -L -o blink.zip https://nosey-dewdrop.github.io/dewfpga/blink.zip && unzip -q blink.zip && rm blink.zip && ls -A
# .vscode Makefile blink.sv blink.xdc blink_tb.sv check_xdc.py
```

file	what it is
<u>blink.sv</u>	the design: LED 0 blinks once a second while switch 0 is on, LED 15 follows switch 0
<u>blink_tb.sv</u>	the testbench: three checks, prints PASS or FAIL
<u>blink.xdc</u>	Digilent's Basys3 master XDC with the 33 pin lines for <code>clk</code> , <code>sw</code> and <code>led</code> uncommented. Vivado reads the same file
<u>check_xdc.py</u>	compares the design's ports with the XDC before place and route, because nextpnr's own error <u>names the wrong port</u>
<u>Makefile</u>	runs the six tools in order. <code>--seed 1</code> makes every build byte-identical
<u>.vscode/tasks.json</u>	<code>Cmd Shift B</code> in VS Code runs <code>make flash</code>

`dewfpga new blink` gives the same `blink.sv`, `blink_tb.sv` and `blink.xdc` without the `Makefile`, and its `tasks.json` says `dewfpga flash` instead.

If you copied the `Makefile` from somewhere else and `make` says `missing separator`, the tabs became spaces: [missing-separator](#).

Plug the board in over USB, switch it on, then from inside the folder:

```
make check
```

```
yosys          /opt/homebrew/bin/yosys
iverilog       /opt/homebrew/bin/iverilog
openFPGALoader /opt/homebrew/bin/openFPGALoader
nextpnr-xilinx /Users/you/fpga/nextpnr-xilinx/build/nextpnr-xilinx
xc7frames2bit  /Users/you/fpga/prjxray/build/tools/xc7frames2bit
chipdb         /Users/you/fpga/chipdb/xc7a35t.bin
fasm2frames    /Users/you/fpga/prjxray/utis/fasm2frames.py
venv           /Users/you/fpga/venv/bin/python
```

Eight lines, each ending in a path, and no –MISSING– anywhere: done. A line such as chipdb –MISSING– names the step of section 3 to redo (3.4 here). Then:

```
make sim      # PASS: 3 checks
make bit      # xdc ok / pnr ok / blink.bit, under five seconds
make flash    # Load SRAM ... done 1
```

The outputs are the same three blocks shown in section 2. `make sim` also writes `blink.vcd`. No waveform viewer is installed by this guide; to look at the waveform, open the [browser testbench](#) and paste the same `blink.sv` and `blink_tb.sv` there, and it draws it.

5. How do you use it for your own lab?

The pin file. The course hands out `Basys3_Master.xdc` with every line commented out. If you do not have it, this puts a copy into the current folder:

```
curl -fsSL -o Basys3_Master.xdc https://nosey-dewdrop.github.io/dewfpga/templates/Basy
s3_Master.xdc
```

Copy it into your lab folder and delete the leading `#` on the `set_property` lines for the pins your module uses. Uncommenting all of them is fine too: pins the design does not use are ignored. The port names in your top module must match the names in that file: `clk`, `sw[0]`, `led[0]`, `seg[0]`, `an[0]`, `btnC`. If your module says `clock`, change the name inside `[get_ports ...]` on that line. The xdc check that runs before place and route lists every port that has no pin and says what to rename.

A design without a clock (assign `led = sw;`) prints `Warning: No clocks found in design and no clocked paths, timing not applicable`; that is correct, not a problem. With the whole file uncommented and no `clk` port, `create_clock: target [get_ports clk] matched nothing` is the same thing said by nextpnr.

With dewfpga: put the `.sv` files and the `.xdc` in one folder, `cd` into it, `dewfpga flash`. With `lab4.sv` and `Basys3_Master.xdc` in the folder that is all. With several `.sv` files, name the top: `dewfpga flash lab4`.

With the Makefile: copy `Makefile` and `check_xdc.py` from section 4 into the lab folder and change the first three lines.

```
TOP := traffic_light
SRCS := traffic_light.sv debounce.sv
XDC := Basys3_Master.xdc
```

TOP is the top module's name, SRCS every .sv in the design, XDC the pin file.

Either way, simulation looks for a testbench (with the Makefile: <top>_tb.sv); the bitstream does not need one. The course's SevSeg_4digit.sv builds as it is; section 7 says what was changed and why.

6. How do you set up VS Code?

Syntax colors and error marks for SystemVerilog:

```
code --install-extension mshr-h.veriloghdl
```

dewfpga new and blink.zip both put a .vscode folder in the project: tasks.json (Cmd Shift B builds and flashes the open design), settings.json (Icarus Verilog as the linter, so a syntax error gets a red underline when you save) and extensions.json (VS Code offers to install the extension above when you open the folder). For a folder you made yourself, copy the .vscode folder out of a dewfpga new project; its three tasks already say dewfpga. On the Makefile path, take the three files from templates/.vscode/ instead; there the tasks say make.

7. Which course file breaks?

One, and it is fixed for you. SevSeg_4digit.sv (also handed out as SevenSegmentDisplay.sv) has a port line that Vivado and Icarus accept and Yosys does not (port-neither-input-nor-output).

```
output [6:0]seg, logic dp,      // dp inherits `output` from the previous port, per the
standard
```

Yosys does not implement that inheritance. dewfpga bit writes the direction in before synthesis and tells you so:

```
note: SevSeg_4digit.sv:4: wrote the port direction in: output [6:0]seg, output logic
dp,      (Yosys needs it; Vivado accepts both)
```

Vivado accepts the changed line too, so the file still works at the lab computer. Two more things Vivado lets through and Yosys refuses came up in old student repos; both stop the build with the line and the fix, because the code is yours to change:

- An instance without a name, trans_3s_clock(clk, reset, out);. The standard requires a name: trans_3s_clock u1(clk, reset, out);.

- A latch in an `always_comb` block: a signal not assigned on every path (an `if` without `else`, a case without `default`). Vivado builds a latch and warns; Yosys refuses. Give the signal a default value at the top of the block.
- A port list where a later port only has a range, `input logic [7:0] d, [2:0] s`: the direction is written in for you, like the seven-segment line above.
- A reset branch that assigns something other than a constant in an `always_ff` with `posedge reset`, and a signal written on both clock edges: both stop with the line and the fix.
- A file Vivado wrote after synthesis (`\<const0>`, `keep_hierarchy`) copied in as if it were source: named, with "take the original .sv".

What else was tried: four modules from old student repos, together in one design. An FSM with `typedef enum`, a 16x8 RAM, a debouncer and the seven-segment driver above: 136 LUTs, 48 flip-flops, timing passes at 154.70 MHz against the 100 MHz clock, bitstream in 4.5 seconds. Language features in those files that both Icarus (`sim`) and Yosys (`bit`) accepted: `parameter`, `generate`, `struct packed`, `$clog2`, `unique case`, packed arrays, interface with `modport`. A submodule file named differently from its module, Turkish characters and a BOM in the source, and spaces in the folder path all went through too. That is four files, not the whole course.

8. What can it not do?

This covers the part of Vivado that CS223 uses, and no more.

- **Nothing from Vivado's IP Catalog.** No Block Design, MicroBlaze, AXI, Clocking Wizard, or the BRAM, VGA and UART cores. 35 old student repos with 240 source files were searched for those; none use any of it. (Searched, not synthesized.)
- **No GUI.** No waveform viewer, schematic or in-chip debugger. `sim` writes a `.vcd`; the [browser testbench](#) draws the waveform for the same files.
- **Two parsers.** Icarus reads your code for simulation, Yosys for synthesis, and they do not support the same SystemVerilog. Code that passes `sim` can fail in `bit`; section 7 is the one case hit so far, and rewriting the line fixed it.
- **The timing number is nextpnr's estimate.** Vivado's report is the official one. In the two designs above the margin was 1.5x (four modules) and 2.8x (blink) over 100 MHz.
- **Unofficial tools.** AMD does not make or support them; `prjxray` worked out the bitstream format without AMD's documentation. If a lab is graded in Vivado, open it in Vivado once before the demo.
- **Memories become distributed RAM** (`-nobraam`). The 16x8 RAM in section 7 mapped to four RAM32M cells. A large memory such as a VGA framebuffer would not fit this way; big multipliers and DSP blocks are untested.
- **One board, one platform.** Basys3 (XC7A35T), Apple Silicon. Intel Mac, Linux and other boards are untested and the installer refuses them.
- **Confirmed on the board** (19 and 20 September): five designs: `blink`, switches to LEDs, the Lab 2 adder/subtractor, the button counter on the display and the traffic-light FSM with sensor hold (the last three are the simulator's examples). The four-module design in section 7 went as far as a bitstream and was not loaded.

- ``ifdef SIM`. The browser simulator defines SIM; the command line does not. A design can use it to shorten a clock divider for the browser and keep the real value for the board, as `blink.sv` does.
-

9. Which versions were tested?

The installer pins `nextpnr-xilinx` and `prjxray` to these commits; section 3 checks out the same ones. Homebrew packages cannot be pinned; the versions below are what was tested.

Yosys	0.69+post (git 143eb14f)
nextpnr-xilinx	0.9.6, openXC7 commit 3fd7878
prjxray	commit c9f02d8; prjxray-db 0.9.1-30-g1768fb3 (artix7)
openFPGALoader	1.1.1
Icarus Verilog	13.0 (stable)
Python	3.14 (Homebrew), venv under ~/fpga
Board	Digilent Basys3, Artix-7 XC7A35T-1CPG236C
Machine	Apple M2, 8 GB RAM, macOS 15 (Darwin 24.2.0)
